Multimodal Mental Health AI: Datasets, Tools & Benchmarks

Comprehensive Directory of Multimodal Corpora, Feature Extractors, Codebases, and Evaluation Protocols

Volume: Volume 3 of 4 | **Focus:** Datasets, Open-Source Toolkits & Evaluation Benchmarks | **Date:** September 2026 | **Synthesized Sources:** 15 Datasets + 13 Toolkits mapped across 8 AI Reviews

1. Master Directory of Multimodal Affective & Mental Health Datasets

A critical finding from the multi-agent review is that while affective computing possesses numerous acted datasets (IEMOCAP, RAVDESS, MELD), true clinical mental health datasets with synchronized facial video and speech are exceedingly rare. Below is the complete directory of 15 key datasets, detailing modalities, sample sizes, access links, licensing, and specific methodological usage rules.

IEMOCAP (2008) — *Text, Video, Audio*

Size / Scale: 12h, 10 actors, ~10k utterances | **Annotations:** 9 emotions + valence/arousal/dominance
License: Research license on request | **Access URL:** Request via https://sail.usc.edu/iemocap/
Agent Mentions (8/8): [ChatGPT] [Claude] [DeepSeek] [Gemini] [GLM] [Kimi] [MiniMax] [Qwen]

Mental Health Usage & Research Discipline: Primary dyadic-conversation benchmark. Pretrain fusion here, but NEVER report it as mental-health evidence (acted English).

MELD (2019) — *Text, Video, Audio*

Size / Scale: ~13k utterances (Friends TV) | **Annotations:** 7 emotions + sentiment
License: Research use | **Access URL:** https://affective-meld.github.io/
Agent Mentions (8/8): [ChatGPT] [Claude] [DeepSeek] [Gemini] [GLM] [Kimi] [MiniMax] [Qwen]

Mental Health Usage & Research Discipline: Multi-party conversation benchmark for DialogueRNN/GCN/COSMIC-style context models. Scripted-TV bias must be disclosed.

CMU-MOSI / CMU-MOSEI (2017/2018) — *Text, Video, Audio*

Size / Scale: MOSI 2.2k clips; MOSEI 23.5k clips | **Annotations:** Sentiment intensity [-3,3] + emotions (MOSEI)
License: Research use | **Access URL:** https://github.com/CMU-MultiComp-Lab/CMU-MultimodalSDK
Agent Mentions (7/8): [ChatGPT] [Claude] [DeepSeek] [Gemini] [GLM] [MiniMax] [Qwen]

Mental Health Usage & Research Discipline: Standard fusion-benchmark pair (MulT, MISA, Self-MM numbers). Opinion-video domain; report MAE/Acc-7/Acc-2/F1 exactly as MERBench prescribes.

RAVDESS (2018) — *Audio, Video (speech+song)*

Size / Scale: 24 actors x 60 clips | **Annotations:** 8 emotions, 2 intensities
License: Research use | **Access URL:** https://www.innovatiana.com/en/datasets/ravdess
Agent Mentions (7/8): [ChatGPT] [Claude] [DeepSeek] [Gemini] [Kimi] [MiniMax] [Qwen]

Mental Health Usage & Research Discipline: Project-named corpus: fine-tune voice encoder here. Acted and clean — always pair with speaker-independent splits (expect ~13% drop vs random splits).

CREMA-D (2014) — *Audio, Video*

Size / Scale: 91 actors, 7,442 clips | **Annotations:** 6 emotions, 4 intensities
License: Research use | **Access URL:** https://github.com/CheyneyComputerScience/CREMA-D
Agent Mentions (5/8): [ChatGPT] [Claude] [DeepSeek] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Project-named corpus: diversity (age/ethnicity) makes it the fairer voice-evaluation set. Combine with RAVDESS for SER training.

FER2013 (2013) — *Static face images (48x48 gray)*

Size / Scale: 35,887 images | **Annotations:** 7 basic emotions
License: Public (Kaggle) | **Access URL:** https://www.kaggle.com/datasets/msambare/fer2013
Agent Mentions (5/8): [ChatGPT] [Claude] [DeepSeek] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Project-named corpus: fine-tune ViT face encoder (expect ~70% acc). Heavy label noise + imbalance — use class-weighted loss; report per-class F1.

AffectNet (2017) — *Static face images (in-the-wild)*

Size / Scale: ~1M images (450k annotated) | **Annotations:** 8 emotions + valence/arousal
License: Research request | **Access URL:** http://mohammadmahoor.com/affectnet/
Agent Mentions (5/8): [ChatGPT] [Claude] [DeepSeek] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Project-named corpus: in-the-wild pretraining for the face encoder; valence/arousal heads seed the dimensional trend output.

DAIC-WOZ / E-DAIC (2014/2019) — *Audio, Video, Text + PHQ-8*

Size / Scale: ~190 interviews | **Annotations:** PHQ-8 depression scores
License: Restricted research agreement | **Access URL:** https://dcapswoz.ict.usc.edu/
Agent Mentions (3/8): [Claude] [Gemini] [MiniMax]

Mental Health Usage & Research Discipline: Clinical bridge: the ONLY listed corpus with validated depression labels. Fine-tune here to transfer basic-emotion features toward distress detection.

ABAW / Aff-Wild2 (2019-2024) — *Face video in-the-wild*

Size / Scale: ~600 videos, ~3M frames | **Annotations:** VA + AUs + 8 expressions
License: Challenge agreement | **Access URL:** https://affective-behavior-analysis-in-the-wild.github.io/8th/
Agent Mentions (2/8): [Claude] [MiniMax]

Mental Health Usage & Research Discipline: Dimensional (valence-arousal) + AU supervision for the trend head; competition protocol is the robustness gold standard.

DFEW (2021) — *Video clips in-the-wild*

Size / Scale: ~16k clips | **Annotations:** 7 emotions
License: Research use | **Access URL:** https://dfew-dataset.github.io/
Agent Mentions (5/8): [ChatGPT] [Claude] [Gemini] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Dynamic FER benchmark for the video face encoder (temporal modelling beyond static FER2013).

MER2023 / MER2024 Challenges (2023/2024) — *Audio, Video, Text*

Size / Scale: Challenge-scale unlabeled + labeled | **Annotations:** Discrete + dimensional + open-vocabulary
License: Challenge terms | **Access URL:** https://github.com/zeroQiaoba/MERTools/tree/master/MER2024
Agent Mentions (6/8): [ChatGPT] [Claude] [Gemini] [GLM] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Adopt MER2024 tracks (incl. open-vocabulary reasoning) for the explainable-head evaluation.

GoEmotions (2020) — *Text (Reddit comments)*

Size / Scale: 58k comments | **Annotations:** 27 emotions + neutral
License: Apache-2.0 | **Access URL:** https://github.com/google-research/google-research/tree/master/goemotions
Agent Mentions (2/8): [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Fine-grained text-emotion labels for the transcript stream; maps basic emotions toward nuanced states.

RAF-DB (2017) — *Static face images*

Size / Scale: ~30k images | **Annotations:** 7 basic + 12 compound emotions
License: Research use | **Access URL:** Research request (W. Deng lab)
Agent Mentions (3/8): [Claude] [Kimi] [MiniMax]

Mental Health Usage & Research Discipline: Cleaner FER2013 alternative for face-encoder benchmarking (POSTER++/DAN numbers reported here).

MSP-Podcast / MSP-IMPROV (2017/2025) — *Audio (+video for IMPROV)*

Size / Scale: 100k+ turns (Podcast) | **Annotations:** Categorical + dimensional
License: Research request | **Access URL:** https://lab-msp.com/MSP/
Agent Mentions (1/8): [Qwen]

Mental Health Usage & Research Discipline: Naturalistic (non-acted) speech-emotion data — closer to monitoring reality than RAVDESS; prioritise for voice generalization tests.

CH-SIMS / SIMS (2020) — *Text, Audio, Video (Chinese)*

Size / Scale: ~2.3k clips | **Annotations:** Multimodal + unimodal sentiment
License: Research use | **Access URL:** Academic request
Agent Mentions (2/8): [Claude] [MiniMax]

Mental Health Usage & Research Discipline: Methodology reference only (unimodal annotation scheme worth copying for the custom dataset); excluded from English benchmarks by the MiniMax filter.

2. Open-Source Toolkits, Frameworks & Implementation Codebases

To avoid reimplementing standard pipelines from scratch, the 8 AI reviews highlighted 13 robust open-source toolkits, SDKs, and reference codebases. These span facial landmark extraction, self-supervised speech modeling, conversational emotion reasoning, and unified benchmarking.

M-SENA platform

Functionality: Integrated MSA/MER platform: data, features, models, evaluation.
Repository / Docs: ACL 2023 system demo (search: M-SENA multimodal sentiment platform)
Agent Mentions (1/8): [GLM]

Role in Proposed System: Experiment harness for face-only / voice-only / fused ablations.

MERTools / MERBench

Functionality: Unified MER data pipeline, fixed splits, standard metrics + challenge toolkit.
Repository / Docs: https://github.com/zeroQiaoba/MERTools
Agent Mentions (2/8): [ChatGPT] [MiniMax]

Role in Proposed System: Adopt its protocol so thesis numbers are comparable across corpora.

CMU-MultimodalSDK

Functionality: Reference SDK for MOSI/MOSEI loading and classic baselines (TFN, MFN, MulT).
Repository / Docs: https://github.com/CMU-MultiComp-Lab/CMU-MultimodalSDK
Agent Mentions (1/8): [MiniMax]

Role in Proposed System: Reproduce baseline fusion numbers before claiming improvements.

S3PRL / SUPERB

Functionality: Self-supervised speech toolkit (wav2vec2, HuBERT, WavLM recipes).
Repository / Docs: https://github.com/s3prl/s3prl
Agent Mentions (1/8): [Claude]

Role in Proposed System: Voice-encoder fine-tuning recipes with frozen-layer schedules.

SpeechBrain

Functionality: Speech toolkit incl. emotion-recognition recipes.
Repository / Docs: https://github.com/speechbrain/speechbrain
Agent Mentions (1/8): [Qwen]

Role in Proposed System: Alternative SER training harness; strong recipe collection.

WhisperX

Functionality: Whisper ASR + word-level timestamps + diarization.
Repository / Docs: https://github.com/m-bain/whisperx
Agent Mentions (2/8): [Gemini] [Qwen]

Role in Proposed System: Transcript stream + utterance segmentation for the streaming pipeline.

FunASR

Functionality: Industrial ASR toolkit (paraformer) with punctuation/timestamps.
Repository / Docs: https://github.com/modelscope/FunASR
Agent Mentions (1/8): [Qwen]

Role in Proposed System: Low-latency ASR alternative for the real-time head.

openSMILE / eGeMAPS

Functionality: Interpretable acoustic feature extractor (88-function eGeMAPS set).
Repository / Docs: https://audeering.github.io/opensmile/
Agent Mentions (3/8): [Claude] [GLM] [Kimi]

Role in Proposed System: MFCC/pitch/spectral baseline features (thesis requirement) + dashboard-grade interpretable trends.

OpenFace 2.0

Functionality: Landmarks, action units, gaze, head pose.
Repository / Docs: https://github.com/TadasBaltrusaitis/OpenFace
Agent Mentions (2/8): [GLM] [Qwen]

Role in Proposed System: AU time-series for explainable reports.

MediaPipe Face

Functionality: 468-point 3D mesh + blendshapes, on-device.
Repository / Docs: https://ai.google.dev/edge/mediapipe
Agent Mentions (1/8): [MiniMax]

Role in Proposed System: Default mobile face front-end (private, real-time).

FER backbones (POSTER / EfficientFace / MAE-DFER / DDAMFN / FaceChannel)

Functionality: Open FER implementations spanning CNN, transformer and masked-autoencoder designs.
Repository / Docs: https://github.com/zczcwh/POSTER ; https://github.com/zengqunzhao/EfficientFace ; https://github.com/sunlicai/MAE-DFER ; https://github.com/SainingZhang/DDAMFN ; https://github.com/pablovin/FaceChannel
Agent Mentions (2/8): [Claude] [MiniMax]

Role in Proposed System: Face-encoder shortlist for ablations.

Conversation-ERC codebases (conv-emotion / COGMEN / MMGCN)

Functionality: Reference DialogueRNN/COSMIC, COGMEN and MMGCN implementations.
Repository / Docs: https://github.com/declare-lab/conv-emotion ; https://github.com/exploration-lab/COGMEN ; https://github.com/hujingwen6666/MMGCN
Agent Mentions (2/8): [Claude] [MiniMax]

Role in Proposed System: Context-modelling references for the session-level mood GRU.

Real-time demo repos

Functionality: mrw-soumik real-time face+voice pipeline; ShahnawazKakarh SSL transfer-learning SER (RAVDESS WF1 0.858 random / 0.728 speaker-independent).
Repository / Docs: GitHub search: mrw-soumik Multimodal-Emotion-Recognition-System ; ShahnawazKakarh speech-emotion-recognition-transfer-learning
Agent Mentions (1/8): **[DeepSeek]**

Role in Proposed System: Starting skeleton for the streaming app + the speaker-independence warning (13% drop) that shapes evaluation.

3. Standardized Evaluation Protocols & Integrity Guidelines

Cross-corpus evaluation in multimodal emotion recognition has historically suffered from split-hacking and over-optimistic validation. To establish publication-grade credibility in the thesis, the following standardized evaluation rules (derived from MERBench and AVEC) must be strictly enforced:

3.1 Speaker-Independent Partitioning (Strict Rule)

Never use random k-fold cross-validation on datasets with repeated speakers (e.g. RAVDESS, CREMA-D, IEMOCAP). DeepSeek and Claude reviews proved that random splits artificially inflate Weighted F1 by ~13-15% because models memorize speaker identity rather than affect. All evaluations must strictly use 5-fold Leave-Speakers-Out (LSO) splits.

3.2 Missing Modality Stress-Testing (MissBench Protocol)

Evaluate model degradation curves across varying missingness ratios (0%, 20%, 50%, 70%) for both audio-missing (camera-only) and video-missing (audio-only) scenarios. Report Modality Inequity Indices (MEI/MLI) to ensure the system does not silently fail on specific sub-populations.

3.3 Confidence Calibration Metrics (ECE & Brier Score)

In health-adjacent monitoring, confidence scores displayed to users must reflect true empirical accuracy. Expected Calibration Error (ECE) must be computed with 15 bins: $ECE = \sum (|B_m| / N) * |acc(B_m) - conf(B_m)|$. Target ECE < 0.05 before release.

3.4 Multimodal vs. Unimodal Ablation Rigor

Every experimental claim must present identical backbone comparisons across: (1) Face-Only, (2) Voice-Only, (3) Transcript-Only, (4) Early Fusion, (5) Late Fusion, and (6) Proposed Cross-Modal Attention (MulT+MISA).